# Practical No: 7

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**Class:-** CO3KB **Subject:-** CGR

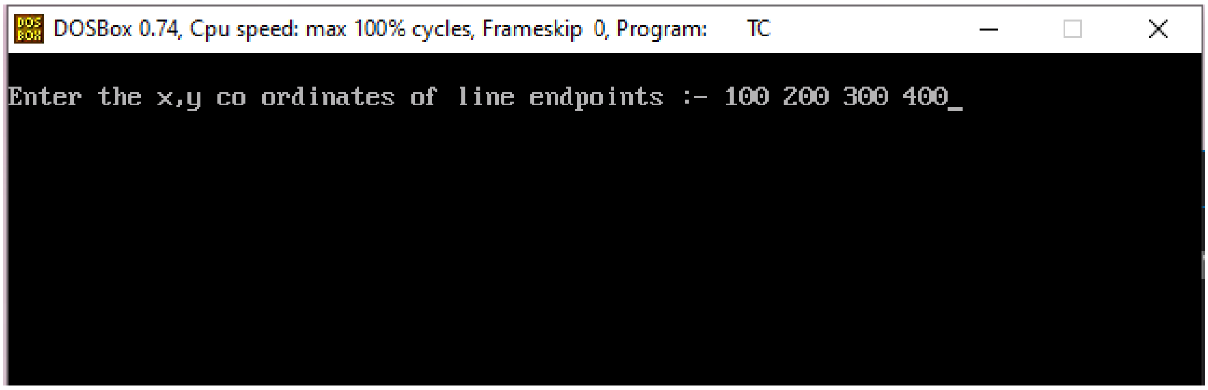
**Aim :- WAP for 2D Translation**

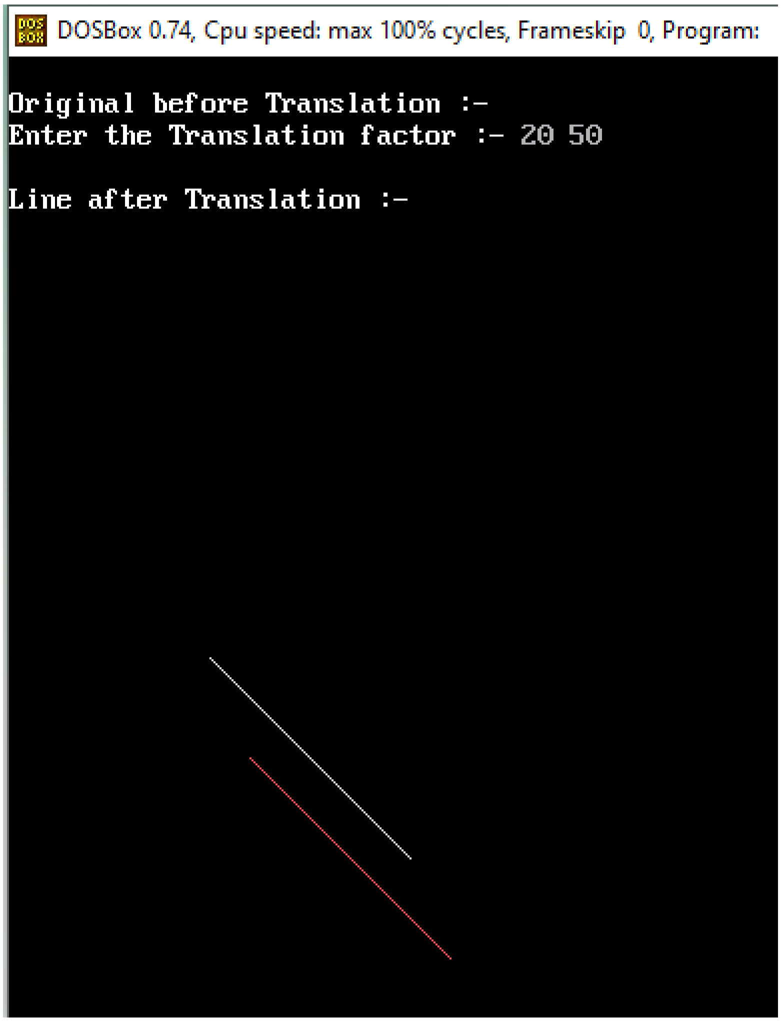
**Program :-**

```
#include<stdio.h>

#include<conio.h>

#include<graphics.h>

void main(){

int gd=DETECT,gm;

int x1,y1,x2,y2;

int tx,ty;

clrscr();

printf("\nEnter the x,y co ordinates of line endpoints :- ");

scanf("%d%d%d%d",&x1,&x2,&y1,&y2);

initgraph(&gd,&gm,"C:\\TurboC3\\BGI");

printf("\nOriginal before Translation :- ");

line(x1,y1,x2,y2);

printf("\nEnter the Translation factor :- ");

scanf("%d%d",&tx,&ty);

x1=x1+tx;

y1=y1+ty;

x2=x2+tx;

y2=y2+ty;

printf("\nLine after Translation :- ");

setcolor(12);

line(x1,y1,x2,y2);
```

```
getch();

closegraph();

}
```

**<u>OUTPUT</u> :-**



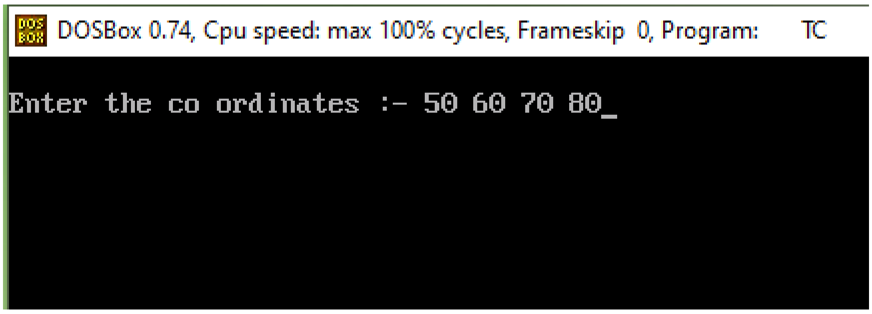



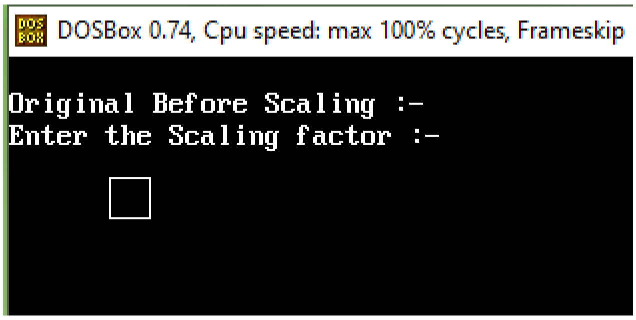
**Aim** :- WAP for 2D Scaling

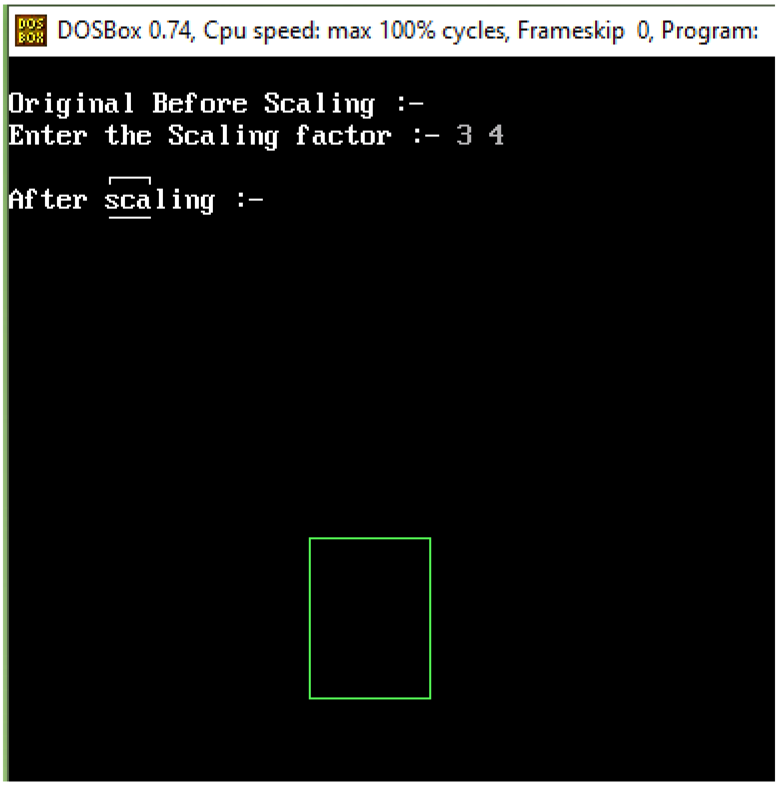
**Program** :-

```
#include<stdio.h>

#include<conio.h>

#include<graphics.h>

void main(){

int gd=DETECT,gm;

int x1,x2,y1,y2;

int Sx,Sy;

clrscr();

printf("\nEnter the co ordinates :- ");

scanf("%d%d%d%d",&x1,&y1,&x2,&y2);

initgraph(&gd,&gm,"C:\\TurboC3\\BGI");

printf("\nOriginal Before Scaling :- ");

rectangle(x1,y1,x2,y2);

printf("\nEnter the Scaling factor :- ");

scanf("%d%d",&Sx,&Sy);

x1=x1*Sx;

y1=y1*Sy;

x2=x2*Sx;

y2=y2*Sy;

printf("\nAfter scaling :- ");

setcolor(10);

rectangle(x1,y1,x2,y2);

getch();

closegraph();

}
```

**OUTPUT :-**


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
Enter the co ordinates :- 50 60 70 80_


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip
Original Before Scaling :-
Enter the Scaling factor :-


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Program:
Original Before Scaling :-
Enter the Scaling factor :- 3 4
After scaling :-